# Hypothesis Testing with E-values - by A. Ramdas and R. Wang

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## Chapter 8 - Handling multiple e-values

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Let $E_1 \dots E_K$ be $K$ e-variable

- Data spliting
- New data collected
- Different e-variable with same data
- ...

We write $\mathcal{K} = [K] = [\![1, K]\!]$

and $\Delta_n := \left\{ (x_1 \dots x_n) \in [0,1]^n , \sum_{k=1}^{n} x_i = 1 \right\}$

## 8.1 Merging e-values under arbitrary dependence

**Def** An e-merging function (of $K$ e-values) is an increasing Borel function

$$F : [0,\infty)^K \longrightarrow [0,\infty)$$

such that for any hypothesis, $F(E_1 \dots E_K)$ is an e-variable for any $E_1 \dots E_K$

- An e-merging function $F$ is symmetric if $F(\vec{e})$ is invariant under any permutation of $\vec{e}$

- An e-merging function $F$ dominates an e-merging function $G$ if $F \geq G$. The domination is strict if $F(\vec{e}) > G(\vec{e})$ for some $\vec{e} \in [0,\infty)^K$

- An e-merging function is admissible if it is not strictly dominated by an e-merging function

- An e-merging function $F$ essentially dominates an e-merging function $G$ if, for all $e \in [0,\infty)^K$

$$G(e) \geq 1 \implies F(e) \geq G(e)$$

ie $F$ is not worse than $G$ when $G$ carries evidence

# Important examples

① Arithmetic mean $M_K$

$$\boxed{M_K(e_1 \dots e_K) = \frac{e_1 + \dots + e_K}{K}}$$

② Weighted arithmetic mean $M_{\vec{\lambda}}$

$$\boxed{M_{\vec{\lambda}}(e_1 \dots e_K) = \sum_{k=1}^{K} \lambda_k e_k + \lambda_{K+1}}$$

for $\vec{\lambda} \in \Delta_{K+1}$

**Prop 8.3** The arithmetic mean $M_K$ essentially dominates any symmetric e-merging function

- In particular if $F$ is
  - an e-merging function
  - positively homogeneous $(F(\lambda \vec{e}) = \lambda F(\vec{e}),\ \lambda > 0)$

Then $F$ is dominated by $M_K$

$\Rightarrow$ geometric mean, harmonic average or minimum are not admissible

- $M_K$ does not dominate every symetric function: for instance $\lambda + (1-\lambda) M_K$ and $M_K$ are non comparable for $\lambda \in (0,1)$

**Th 8.4** For a function $F : \mathbb{R}_+^K \to \mathbb{R}_+$

- if $F$ is an e-merging function, then $F \leqslant M_{\vec{\lambda}}$ for some $\vec{\lambda} \in \Delta_{K+1}$
- $F$ is an admissible e-merging function iff $F = M_{\vec{\lambda}}$ for some $\vec{\lambda} \in \Delta_{K+1}$

## 8.2 Independent e-values and their products

**Def 8.5** An ie-merging function is a Borel function $F : [0,\infty)^K \to [0,\infty)$ s.t. for any hypothesis $F(E_1 \dots E_k)$ is an e variable for any independent $E_1 \dots E_k$

- An ie-merging function $F$ weakly dominates an ie-merging function $G$ if for all $e_1 \dots e_K$
$$(e_1 \dots e_k) \in [1,\infty)^K \Rightarrow F(e_1 \dots e_K) \geq G(e_1 \dots e_k)$$
- The corresponding definition of domination, strict domination and admissibility can be adapted

# Important examples

③ The product $\Pi_K$

$$\boxed{\Pi_K(e_1 \dots e_K) = \prod_{k=1}^{K} e_k}$$

④ The U-statistics function $U_n$

$$\boxed{U_n(e_1 \dots e_K) = \frac{1}{\binom{K}{n}} \sum_{\substack{A \subseteq \mathcal{K} \\ |A| = n}} \left( \prod_{k \in A} e_k \right)} \quad n \in \mathcal{K}$$

- $\Pi_K$, $M_K$ and $1$ are U statistics
- Convexe mixture of U-statistics are ie-merging funct

**Prop 8.6** Fix an atomless probability measure $\mathbb{P}$. For an increasing Borel function $F : [0,\infty)^K \to [0,\infty)$, if $\mathbb{E}^{\mathbb{P}}[F(\vec{E})] = 1$ for all vectors $\vec{E}$ of exact e-variable [resp. for all vectors $\vec{E}$ of indpdt e-variable] then $F$ is an admissible e-merging function [resp. admissible ie-merging function]

In particular, U-statistics function and their convex mixture are admissible ie-merging function.

- **Prop 8.7** The product ie-merging function $\Pi_K$ weakly dominates any ie-merging function

- The convex mixture of U-statistics function does not form a complete class of admissible ie-merging function. For instance

$$f(e_1, e_2) = \frac{1}{2}\left(\frac{e_1}{1+e_1} + \frac{e_2}{1+e_2}\right)(1 + e_1 e_2)$$

is an admissible ie-merging function which is not a convex mixture of U-statistics.

## 8.3 Sequential e-values and martingale merging functions

Reminder

**Def 7.19** The e-variable $E_t$ with $t \in \mathbb{T}_+$ for $\mathcal{P}$, adapted to the filtration $\mathcal{F}$, are sequential if $\mathbb{E}^{P}[E_t \mid \mathcal{F}_{t-1}] \leq 1$ for $t \in \mathbb{T}_+$ and $\mathbb{P} \in \mathcal{P}$.
In case the filtration $\mathcal{F}$ is not specified, by default we mean the natural filtration
$\mathcal{F}_t = \sigma(E_1 \ldots E_t)$

In particular one can take $\mathbb{T}_+ = \{1 \ldots K\}$

**Def 8.9** An ==se-merging function== is a Borel function $F: [0,\infty)^K \to [0,\infty)$ st for any hypothesis $F(E_1 \dots E_K)$ is an e-variable for any sequential e-variable $E_1 \dots E_K$

- An se-merging function $F$ is ==exact== if $F(E_1 \dots E_K)$ is an exact e-variable for all sequential exact e-variable $E_1 \dots E_K$
- The corresponding definition of ==domination==, ==strict domination== and ==admissibility== can be adapted

__NB__ se-merging $\Longrightarrow$ ie-merging

$\not\Longleftarrow$ (remember the 🥦)

# Important examples

(5) All convex mixture of U-statistics function are admissible se-merging function

(6) Martingale merging function:

**Def 8.10** A function $F : [0,\infty)^K \to [0,\infty]$ is a martingale merging function if

$$F(e_1 \dots e_K) = \prod_{k=1}^{K} (1 - \lambda_k + \lambda_k e_k$$

where $\lambda_k = \lambda_k (e_1 \dots e_{k-1})$ for $k \geq 2$ is in $[0,1]$ and $\lambda_1 \in [0,1]$ is a constant.

Made with Goodnotes

**Prop 8.11** Any martingale merging function is an exact se-merging function. Moreover, a convex mixture of martingale merging function is again a martingale merging function

**Th 8.12** Any se-merging function is dominated by a martingale merging function.

## Examples

- This contains but is larger than the U- statistics functions
- Hit and stop for a fixed $\alpha \in (0,1)$ and for each $k \in \mathcal{K}$, if $F(e_1 \dots e_k, 1 \dots 1) \geq \frac{1}{\alpha}$ then $\lambda_j = 0$ for all $j \geq k+1$ otherwise $\lambda_{k+1} > 0$
- Empirically adapted e- process

$$\lambda_k \in \underset{\lambda \in [0,\gamma]}{\operatorname{argmax}} \frac{1}{k-1} \sum_{s=1}^{k-1} \log((1-\lambda) + \lambda e_s)$$

for $k \geq 1$ and $\lambda_1 = 0$

for $\gamma \in [0,1]$

## 8.4 Mean - Variance trade - off

**Prop 8.16** Let $F : [0,\infty)^K \to [0,\infty)$ be an se-merging function. For indpdt, non negative r.v $E_1 \dots E_K$ under $Q$ with $E^Q[E_k] \geq 1$ $\forall k \in \mathcal{K}$ (ie $E$ powered against $Q$), we have for every $m \in [1,\infty)$

$$\mathbb{E}^Q[F(E_1 \dots E_K)^m] \leq \prod_{k=1}^K \mathbb{E}^Q[E_k^m] = \mathbb{E}^Q[\Pi_k(E_1..E_K)^m]$$

In particular, if $F$ is exact and $E_1 \dots E_K$ are indpdt e-variable for $\mathbb{P}$, then

$$\mathrm{Var}^{\mathbb{P}}(F(E_1 \dots E_k)) \leq \mathrm{Var}^{\mathbb{P}}(\Pi_K(E_1 \dots E_k))$$

# Takeaway

- merging arbitrarily dependent e-values
  - ↳ arithmetic mean $M_K$ or weighted arithmetic mean $M_{\vec{\lambda}}$
- merging sequential e-values
  - ↳ martingale merging function
    - ↳ by default we empirically adaptive
- merging indpdt e-values
  - ↳ any U-statistics function
    - ↳ Attention to the mean-varicnce trade off!
- merging iid e-values
  - ↳ empirically adaptive martingale merging function has asymptotic log optimality